

L2 – 3.3 Quotient of Linear Functions

MHF4U

Jensen

Part 1: Key Features of the Quotient of Linear Functions

Features of $f(x) = \frac{ax+b}{cx+d}$

- If an x value is a zero of the denominator ONLY, this results in a vertical asymptote
 - Equation of vertical asymptote is $x = \frac{-d}{c}$
- If an x value is a zero of the numerator AND denominator, this results in a **hole** in the graph NOT a vertical asymptote
- There is a horizontal asymptote at the ratio of the leading coefficients
 - Equation of horizontal asymptote is $y = \frac{a}{c}$
- Forms a **Hyperbola**: the two branches of the graph of the function are equidistant from the point of intersection of the vertical and horizontal asymptotes
 - Once you know the shape of one branch, you can translate the points to graph the other branch
- You can find the x -intercept by setting $y = 0$ and solving for x
 - This results in $\left(\frac{-b}{a}, 0\right)$
- You can find the y -intercept by setting $x = 0$ and solving for y
 - This results in $\left(0, \frac{b}{d}\right)$

Part 2: Graphing a Quotient of Linear Functions

Example 1: Graph each of the following functions

a) $f(x) = \frac{x-3}{x+2}$

VA: $x+2=0$
 $x=-2$

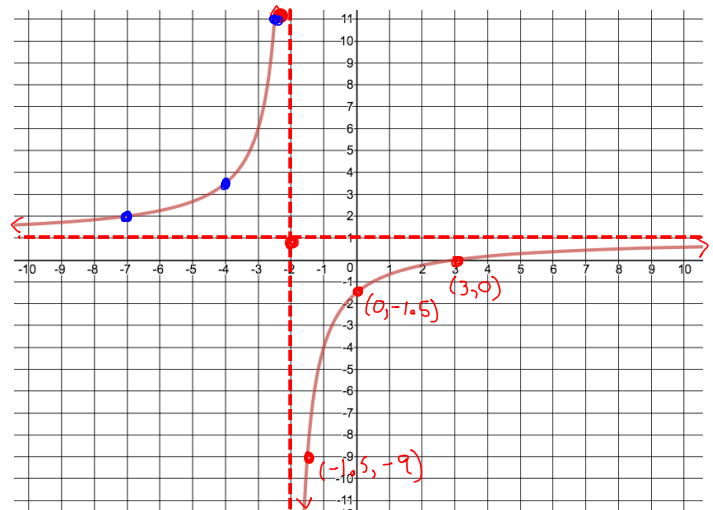
HA: $\frac{1}{1} = 1$
 $y=1$

x -int: $0 = \frac{x-3}{x+2}$
 $0 = x-3$
 $x=3$
 $(3, 0)$

y -int: $f(0) = \frac{0-3}{0+2}$
 $= -\frac{3}{2}$
 $(0, -1.5)$

Another point:

$f(-1.5) = \frac{-1.5-3}{-1.5+2}$
 $= \frac{-4.5}{0.5}$
 $= -9$



$$b) g(x) = \frac{2x-3}{x-1}$$

$$\text{VA: } x-1=0 \\ x=1$$

$$\text{HA: } \frac{2}{1}=2 \\ y=2$$

$$\begin{aligned} \text{x-int: } 0 &= \frac{2x-3}{x-1} \\ 0 &= 2x-3 \\ x &= \frac{3}{2} \\ (1.5, 0) \end{aligned}$$

$$\begin{aligned} \text{y-int: } f(0) &= \frac{2(0)-3}{0-1} \\ &= 3 \\ (0, 3) \end{aligned}$$

Other points:

$$\begin{aligned} f(2) &= \frac{2(2)-3}{2-1} = \frac{1}{1} = 1 \\ (2, 1) \end{aligned}$$

$$\begin{aligned} f(3) &= \frac{2(3)-3}{3-1} = \frac{3}{2} \\ (3, 1.5) \end{aligned}$$

